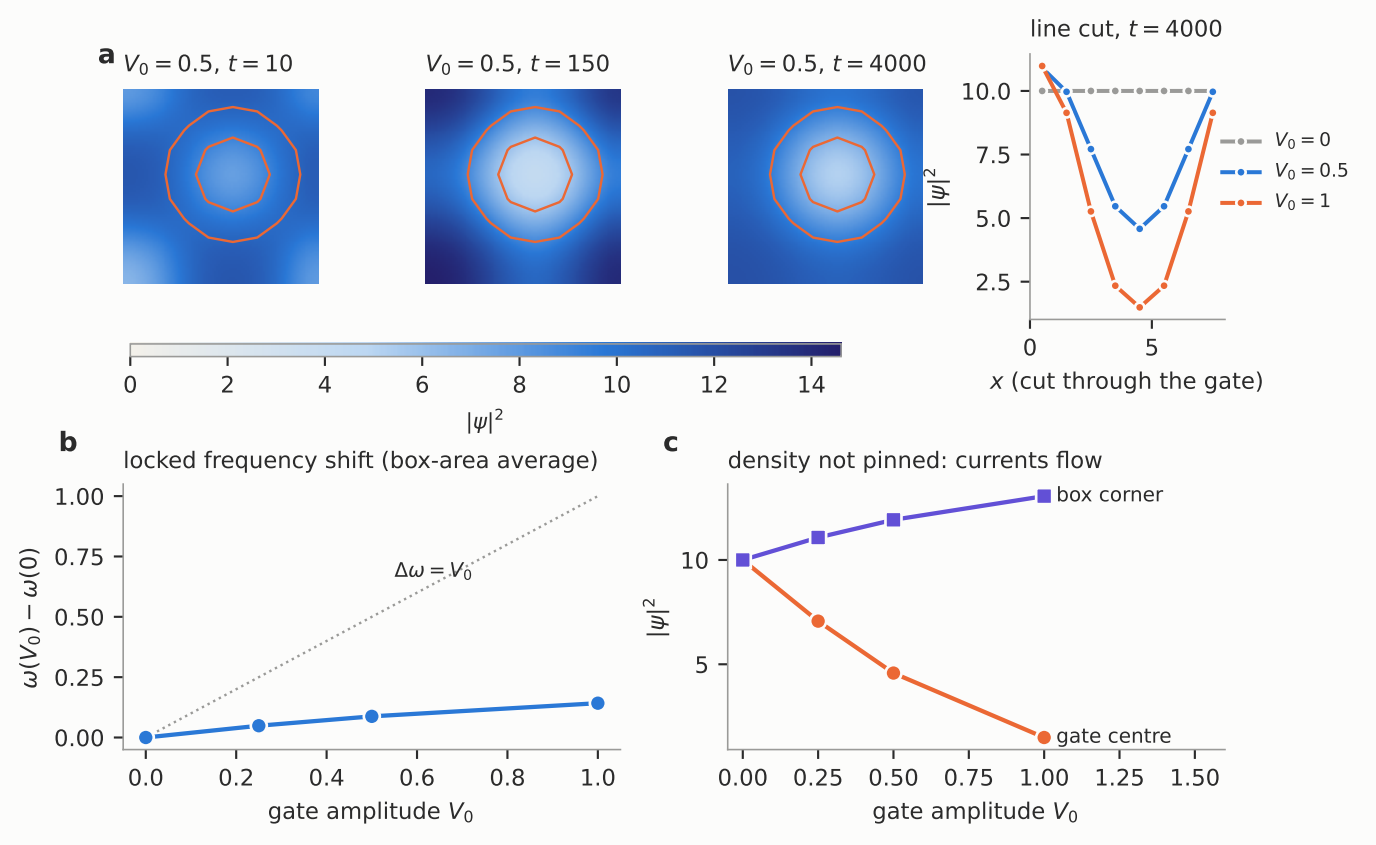
a
$V_0 = 0.5,\ t = 10$
$V_0 = 0.5,\ t = 150$
$V_0 = 0.5,\ t = 4000$
0
2
4
6
8
10
12
14
$|\psi|^2$
line cut, $t = 4000$
10.0
7.5
5.0
2.5
$|\psi|^2$
0
5
x (cut through the gate)
$V_0 = 0$
$V_0 = 0.5$
$V_0 = 1$
b
locked frequency shift (box-area average)
1.00
0.75
0.50
0.25
0.00
$\omega(V_0) - \omega(0)$
$\Delta\omega = V_0$
0.0
0.2
0.4
0.6
0.8
1.0
gate amplitude V_0
c
density not pinned: currents flow
box corner
gate centre
10
5
$|\psi|^2$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
gate amplitude V_0